Gürkan Doğukan Karakurt

GRADUATE RESEARCHER · CIVIL ENGINEER · SOFTWARE DEVELOPER

ORCID: 0009-0008-3277-5405 Google Scholar LinkedIn

MSc candidate and Research Assistant in Construction Engineering & Management at METU, supervised by Prof. Dr. Rıfat Sönmez. Thesis applies graph neural networks to BIM models to generate complete construction precedence schedules from a building's geometry and standardized activity classifications. Background in civil engineering (BSc, METU) with a minor in Corporate Finance and a long-running practice in full-stack software development and graphic design.

EDUCATION

2025 – Present

MSc, Construction Engineering & Management

Middle East Technical University (METU), Ankara

Research Assistant. Thesis on BIM-driven scheduling with graph neural networks, supervised by Prof. Dr. Rıfat Sönmez. Transferred from the Geotechnical Engineering programme in Spring 2025.

2017 – 2023

BSc, Civil Engineering

Middle East Technical University (METU) — GPA 3.35 / 4.00

Coursework in structural analysis, geotechnics, hydraulics, construction management, and Building Information Modeling.

2019 – 2023

Minor, Corporate Finance

Middle East Technical University (METU) — GPA 3.39 / 4.00

Minor programme covering financial analysis, valuation, and quantitative methods alongside the civil engineering degree.

RESEARCH

Predicting Construction Precedence Relationships from BIM Geometry with Machine Learning

I am building a graph neural network that reads a BIM model with assigned OmniClass activities and outputs the full construction schedule — every FS, SS, FF, SF precedence link — so that schedules regenerate themselves whenever the design changes.

PROBLEM

BIM and the construction schedule still do not talk to each other. Planners rebuild precedence by hand for every project, and recent text-only ML approaches ignore geometry and break down at real project scale.

APPROACH

A single GNN (GraphSAGE / GAT) consumes a three-layer IFC representation — native IFC class and geometry, Uniformat II, and OmniClass T22 — and passes messages over BIM topology to score every candidate precedence link in the schedule.

CONTRIBUTION

First end-to-end whole-schedule precedence predictor grounded in BIM topology and standardized classification codes instead of free-form activity text. Designed to transfer across firms and projects.

EXPERIENCE

Oct 2025 – Present

Research Assistant — Construction Engineering & Management

Middle East Technical University (METU), Department of Civil Engineering

Graduate research and teaching support within the Construction Engineering and Management division. Focus areas include BIM, graph neural networks for construction scheduling, and machine learning applied to the built environment.

Dec 2024 – Oct 2025

Software Engineer

Mega Mühendislik, Türkiye

Developed software for engineering workflows, contributing across the stack to ship reliable internal tooling before moving full-time into graduate research at METU.

2023 – Present

Founder & Full-Stack Developer — Diyet Cebimde

Self-initiated · AI-assisted diet & nutrition platform

Solo-built mobile platform that generates AI-driven weekly diet plans from each user's profile and goals, with photo-based meal recognition, manual tracking, weight/measurement logging, and a memory-equipped chatbot. Hybrid pipeline (Gemini for meal selection, PuLP linear programming for portion optimisation) and snapshot architecture keep historical plans stable across catalog updates. Three repositories: Django + DRF + PostgreSQL backend, React 18 + Vite admin panel, Expo + React Native mobile client.

2023 – 2024

Head of Quantitative Strategies

Vortex · Remote (Georgia)

Designed and developed algorithmic trading systems for financial markets — scalping and swing strategies, performance tuning, and per-investor mandate customisation.

Summer 2022

Civil Engineering Intern

ES Group — Erdemir Port Project, Zonguldak

Worked across site operations, planning, and progress payments on a port construction project. Contributed to steel-pile design and took responsibility in cost control and project scheduling.

2020 – 2024

Freelance Graphic Designer · 2024 Local Elections Lead Designer

Independent

Delivered posters, brochures, social media content, and campaign materials across freelance projects. Served as lead designer for a 2024 Turkish mayoral campaign, owning the full digital content production pipeline.

SKILLS

Research & Engineering: BIM (Revit, IFC), Construction Mgmt (scheduling, CPM), Graph Neural Networks (GraphSAGE / GAT), Machine Learning (PyTorch), AutoCAD (upper-intermediate), SAP2000 (beginner-intermediate), EPANET (advanced), MS Project (intermediate)

Software Development: Python & Django (advanced), HTML / CSS / JS (advanced), React (intermediate), PostgreSQL (intermediate), MATLAB / VBA (advanced)

Design: Photoshop (advanced), Illustrator (advanced), Premiere Pro (advanced), After Effects (intermediate)

Languages: Turkish (native), English (C1)

Personal contact details (email, phone, address) are intentionally omitted from this CV for privacy. Please reach out through LinkedIn or ORCID for academic correspondence.